$$\int_1^\pi \frac{\cos(\ln x)}{x^2} dx$$

$$u := \ln x$$

$$du = \frac{1}{x} dx$$

$$\int_0^{\ln \pi} \frac{\cos u}{e^u} du$$

$$w := \cos u \ dv := \frac{1}{e^u} du$$

$$dw = -\sin u du \ v = -\frac{1}{e^u}$$

$$-\frac{\cos u}{e^u} \ |_0^{\ln \pi} - \int_0^{\ln \pi} \frac{\sin u}{e^u} du$$

$$t := \sin u$$

$$dt = \cos u du$$

$$-\frac{\cos(\ln \pi)}{\pi} + 1 - \left(-\frac{\sin u}{e^u} \ |_0^{\ln \pi} + \int_0^{\ln \pi} \frac{\cos u}{e^u} du\right) = \int_0^{\ln \pi} \frac{\cos u}{e^u} du$$

$$\int_0^{\ln \pi} \frac{\cos u}{e^u} du = -\frac{\cos(\ln \pi)}{\pi} + 1 + \frac{\sin u}{e^u} \ |_0^{\ln \pi} - \int_0^{\ln \pi} \frac{\cos u}{e^u} du$$

$$2 \int_0^{\ln \pi} \frac{\cos u}{e^u} du = -\frac{\cos(\ln \pi)}{\pi} + 1 + \frac{\sin(\ln \pi)}{\pi}$$

$$\int_0^{\ln \pi} \frac{\cos u}{e^u} du = \frac{-\frac{\cos(\ln \pi)}{\pi} + \frac{\sin(\ln \pi)}{\pi} + 1}{2}$$